



Digital Communication IC Design

Lecture 2: Modulation Coding Scheme, Optimal Detection and BER

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Outline

- ☐ Modulation Coding Scheme
- ☐ Optimal Detection and BER

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Pulse Amplitude Modulation (PAM)

- ❑ A sequence of a basic pulse is amplitude-modulated by a sequence of data symbols.
 - Baseband PAM: wireline transmission
 - Passband PAM: wireless transmission
 - Special cases of passband PAM: Phase-shift keying (PSK), amplitude and phase modulation (AM-PM) and quadrature amplitude modulation (QAM)



PAM – Coder and Bit/Symbol-Rate

- ❑ Encoder
 - Convert the incoming bit stream into a stream of symbols
 - A bit of incoming bit stream $\in [0, 1]$ and a symbol $\in [-1, 1]$ for PAM-2,
 - A bit of incoming bit stream $\in [0, 1]$ with coding to form a symbol $\in [-1, 0, 1]$ for PAM-3,
 - Two bits of incoming bit stream $\in [00, 01, 10, 11]$ a symbol $\in [-3, -1, 1, 3]$ for PAM-4
- ❑ Bit rate vs. symbol rate
 - For PAM-2, bit-rate = symbol-rate (or sample-rate)
 - For PAM-4, bit-rate = 2 symbol-rate

Example – Automotive Ethernet

□ IEEE 802.3bp -- 3B2T

■ 3 input bits as a group {B[0], B[1], B[2]}

- B[0]: the first bit
- B[1]: the second bit
- B[2]: the third bit

B[2], B[1], B[0]	T[1], T[0]
000	-1,-1
001	0,-1
010	-1,0
011	-1,+1
100	+1,0
101	+1,-1
110	+1,+1
111	0,+1

■ 2 output bits as the transmitted symbol {T[0], T[1]}

- T[0]: the first transmitted symbol (or sample)
- T[1]: the second transmitted symbol (or sample)

■ PAM-3 with code rate of 3/2

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Binary Phase-Shift Keying (BPSK)

□ Constellation in 2-D coordinate system

■ Signal $s_1(t)$

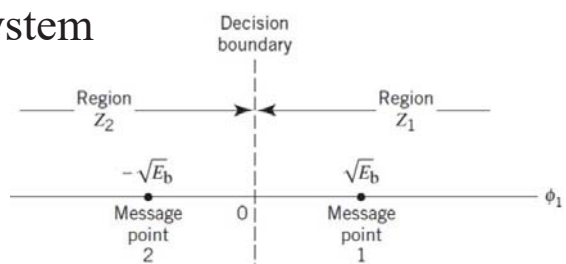
$$s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cdot \cos(2\pi f_c t), 0 \leq t \leq T_b$$

■ Signal $s_2(t)$

$$s_2(t) = \sqrt{\frac{2E_b}{T_b}} \cdot \cos(2\pi f_c t + \pi) = -\sqrt{\frac{2E_b}{T_b}} \cdot \cos(2\pi f_c t), 0 \leq t \leq T_b$$

■ $\phi_1(t) = \sqrt{\frac{2}{T_b}} \cdot \cos(2\pi f_c t), 0 \leq t \leq T_b$

- T_b : the bit duration; (or bit rate, symbol rate)
- E_b : the transmitted signal energy per bit;



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Binary Phase-Shift Keying (cont'd)

□ The transmitted signal $s_1(t)$ and $s_2(t)$

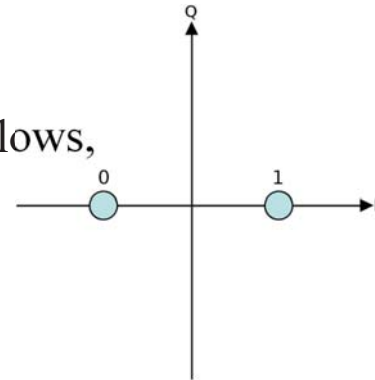
■ $s_1(t) = \sqrt{E_b} \phi_1(t)$

■ $s_2(t) = -\sqrt{E_b} \phi_1(t)$

□ In general, the coding of BPSK is as follows,

■ Input bit stream: b_n

■ Output symbol (signal): s_n



□ Average signal power

■ P_{avg}

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4-PSK or Quadrature PSK (QPSK)

□ Signal constellation $\{-1, -j, +1, +j\}$

□ The phase for 4-PSK

■ $\theta_m = \left\{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\right\}$

■ Also called quadrature PSK (QPSK)

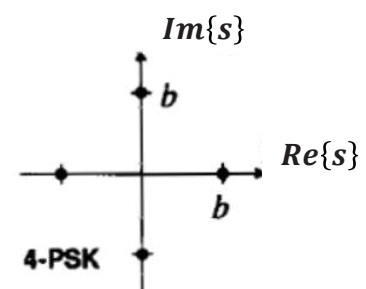
■ $M = 4$ and each symbol represented $\log_2 M = 2$ bits

□ The transmitted signal

$s(t) = s_1(t)\cos(w_c t) - s_2(t)\sin(w_c t)$

■ $s_1(t) = b \cdot \sum_{m=-\infty}^{\infty} \cos(\theta_m)g(t - mT)$

■ $s_2(t) = b \cdot \sum_{m=-\infty}^{\infty} \sin(\theta_m)g(t - mT)$



Alternative QPSK

- The signal model is defined as

$$s_m(t) = A \cdot \cos(w_c t + \theta_m), 0 \leq t \leq T, \quad m = 1, 2, 3, 4$$

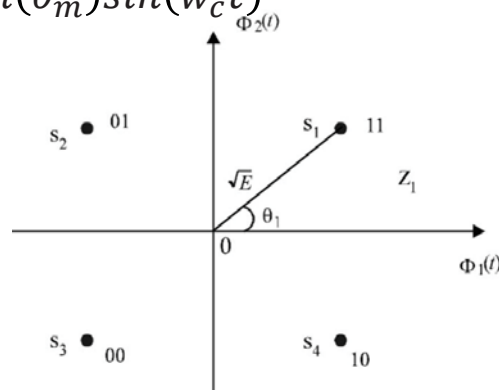
where $\theta_m = (2m - 1)\pi/4$

- $\theta_m = \left\{ \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \right\}$; T : the symbol duration

- The signal model can be revised as

$$\begin{aligned} s_m(t) &= \sqrt{E} \cos(\theta_m) \cos(w_c t) - \sqrt{E} \sin(\theta_m) \sin(w_c t) \\ &= s_{m1} \phi_1(t) + s_{m2} \phi_2(t) \end{aligned}$$

- $s_{m1} = \sqrt{E} \cos \theta_m$
- $s_{m2} = \sqrt{E} \sin \theta_m$
- $\theta_m = \tan^{-1}(s_{m2}/s_{m1})$
- $\phi_1(t) = \sqrt{2/T} \cdot \cos(2\pi f_c t), 0 \leq t \leq T$
- $\phi_2(t) = \sqrt{2/T} \cdot \sin(2\pi f_c t), 0 \leq t \leq T$



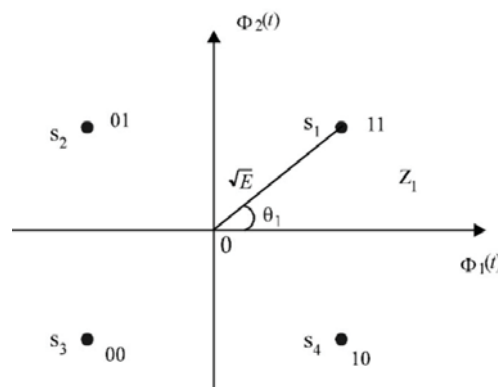
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Alternative QPSK (cont'd)

- Signal space characterization

Gray-encoded input dibit	Phase of QPSK signal (radians)	Coordinates of message points	
		s_{i1}	s_{i2}
11	$\pi/4$	$+\sqrt{E/2}$	$+\sqrt{E/2}$
01	$3\pi/4$	$-\sqrt{E/2}$	$+\sqrt{E/2}$
00	$5\pi/4$	$-\sqrt{E/2}$	$-\sqrt{E/2}$
10	$7\pi/4$	$+\sqrt{E/2}$	$-\sqrt{E/2}$



- Average signal power

- P_{avg}

M-ARY PSK (MPSK)

- ❑ Increase the BW efficiency of PSK modulation
- ❑ M-ary PSK signal is defined as

$$s_m(t) = A \cdot \cos(w_c t + \theta_m), 0 \leq t \leq T, \quad m = 1, 2, \dots, M$$

where $\theta_m = (2m - 1)\pi/M$

- M : the power of 2 ($M = 2^n$ and $n = \log_2 M$)

- ❑ The $s_m(t)$ can be revised as

$$\begin{aligned} s_m(t) &= A \cdot \cos\theta_m \cos(w_c t) - A \cdot \sin\theta_m \sin(w_c t) \\ &= s_{m1}\phi_1(t) - s_{m2}\phi_2(t) \end{aligned}$$

where $\phi_1(t)$ and $\phi_2(t)$ are orthonormal basic function.

- $\phi_1(t) = \sqrt{2/T} \cdot \cos(2\pi f_c t), 0 \leq t \leq T$
- $\phi_2(t) = \sqrt{2/T} \cdot \sin(2\pi f_c t), 0 \leq t \leq T$

M-ARY PSK (cont'd)

- ❑ Both s_{m1} and s_{m2} are given as

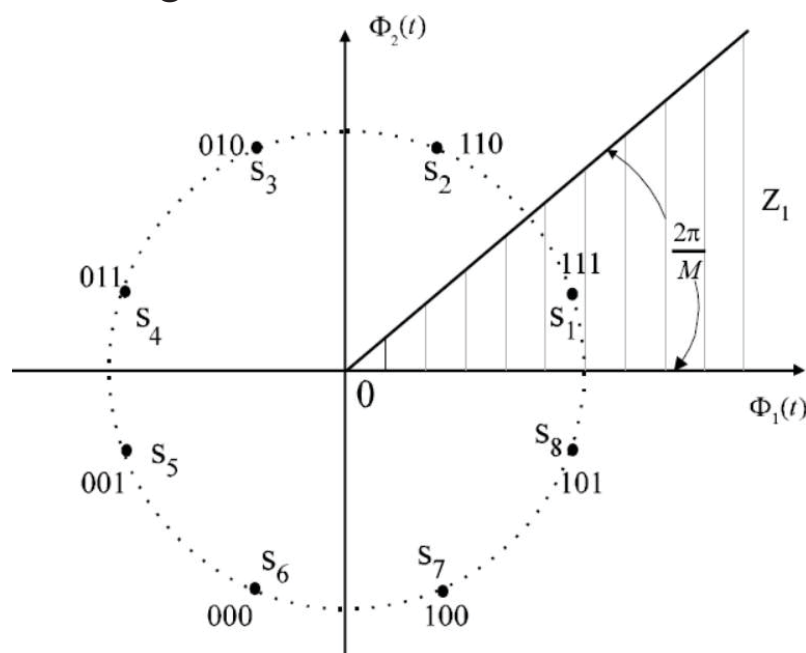
$$s_{m1} = \int_0^T s_m(t)\phi_1(t)dt = \sqrt{E}\cos\theta_m$$

$$s_{m2} = \int_0^T s_m(t)\phi_2(t)dt = \sqrt{E}\sin\theta_m$$

- $E = \frac{A^2 T}{2}$: the symbol energy of the signal (A: the amplitude)
- $\theta_m = \tan^{-1}(s_{m2}/s_{m1})$
- (s_{m1}, s_{m2}) : the m th constellation point
- $(\sqrt{E}, \theta_m)$: the polar coordinates of the signal
 - $\sqrt{E}$ (magnitude), θ_m (phase)
- Gray coding used in the signal assignment in MPSK
 - One-bit difference to two adjacent signal in the constellation

M-ARY PSK (cont'd)

- Gray coded bit assignment for 8-PSK constellation



Fuqin Xiong, 2006

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M-ARY PSK (cont'd)

- On the entire time axis, the MPSK can be expressed as $s(t) = s_1(t)\cos(w_c t) - s_2(t)\sin(w_c t)$, $-\infty < t < \infty$
where

$$s_1(t) = A \sum_{m=-\infty}^{\infty} \cos(\theta_m) g(t - mT)$$

$$s_2(t) = A \sum_{m=-\infty}^{\infty} \sin(\theta_m) g(t - mT)$$

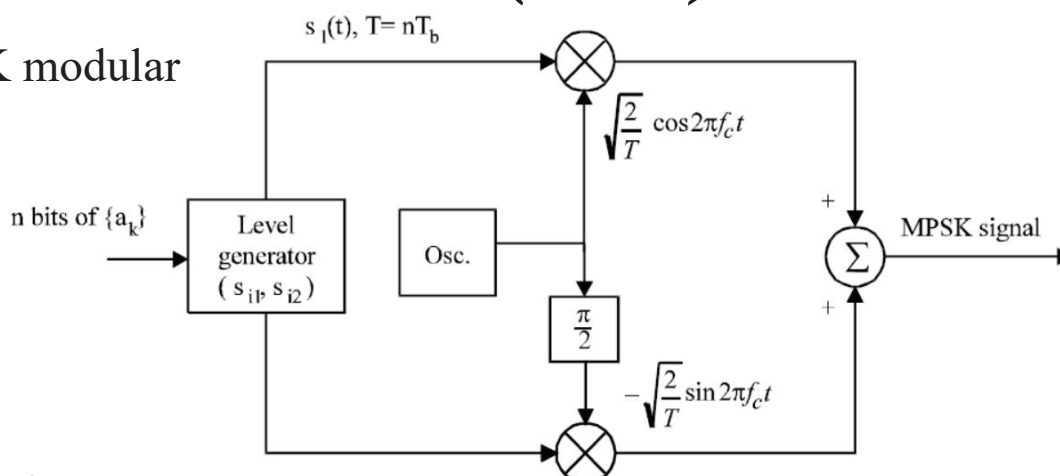
- θ_m is one of M phases determined by the input binary bit
- $g(t)$: the rectangular pulse with unit amplitude defined on $[0, T]$

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M-ARY PSK (cont'd)

❑ MPSK modular



- Level generator: $s_2(t), T = nT_b$
- Each n-tuple (n constellation points) of the input bits used to control the level generator
 - Serial-to-parallel conversion
 - T_b : the bit period (duration)
 - T : the symbol period (duration)

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Quadrature Amplitude Modulation (QAM)

- ❑ Achieve higher BW efficiency than MPSK with the same average signal power
- ❑ The signal model of QAM

$$s_m(t) = A_m \cdot \cos(\omega_c t + \theta_m), \quad m = 1, 2, \dots, M$$
 - A_m : the amplitude of m th signal M-ary QAM scheme
 - θ_m : the phase of m th signal M-ary QAM scheme
- ❑ The signal model of QAM with pulse-shaping filter

$$s_m(t) = A_m \cdot p(t) \cos(\omega_c t + \theta_m), \quad m = 1, 2, \dots, M$$
 - $p(t)$: the smooth pulse defined in $[0, T]$

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QAM (cont'd)

- The signal model of QAM with pulse-shaping filter can be revised as

$$s_m(t) = A_{m1} \cdot p(t)\cos(w_c t) - A_{m2} \cdot p(t)\sin(w_c t)$$

- $A_{m1} = A_m \cos(\theta_m)$ and $A_{m2} = A_m \sin(\theta_m)$

- $A_m = \sqrt{A_{m1}^2 + A_{m2}^2}$

- The QAM signal can be expressed as a linear combination two orthogonal function.

$$s_m(t) = s_{m1}\phi_1(t) + s_{m2}\phi_2(t)$$

- $\phi_1(t) = \sqrt{2/E_p} p(t)\cos(w_c t)$ and $\phi_2(t) = \sqrt{2/E_p} p(t)\sin(w_c t)$

- E_p : the energy of $p(t)$, $E_p = \int_0^T p^2(t)dt$

- $\sqrt{2/E_p}$: used to normalize the basis function $\phi_1(t)$ and $\phi_2(t)$

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QAM (cont'd)

- $s_{m1} = \sqrt{E_p/2} A_{m1} = \sqrt{E_p/2} A_m \cos(\theta_m)$

- $s_{m2} = \sqrt{E_p/2} A_{m2} = \sqrt{E_p/2} A_m \sin(\theta_m)$

- A QAM signal

- Represented by a point (or vector, or phasor) with coordinates (s_{m1}, s_{m2})

- Alternative QAM signal

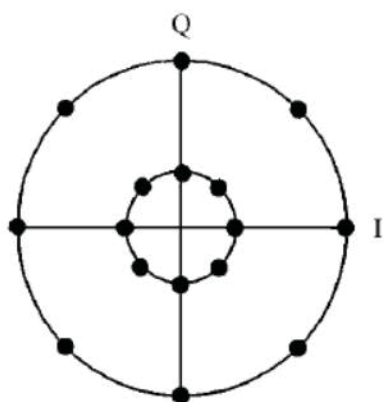
- Two axes $p(t)\cos(w_c t)$ and $-p(t)\sin(w_c t)$ denoted as I-axis and Q-axis

- Signal coordinates (A_{m1}, A_{m2})

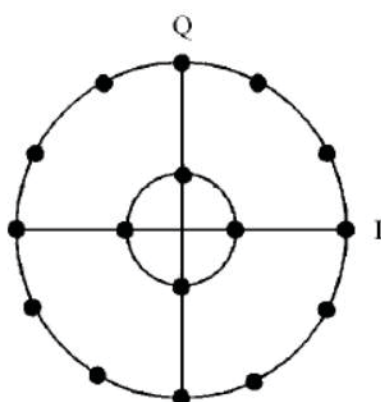
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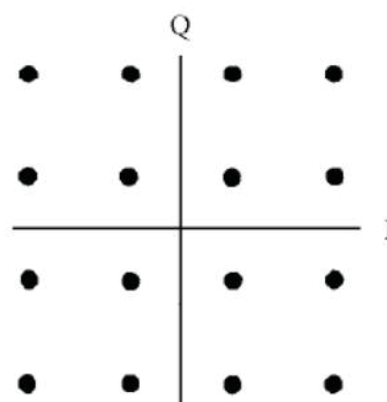
QAM Constellation



Type I QAM constellation



Type II QAM constellation

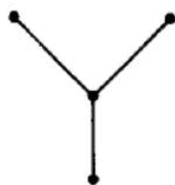


Type III QAM constellation

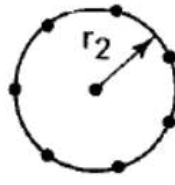
Various QAM Constellations



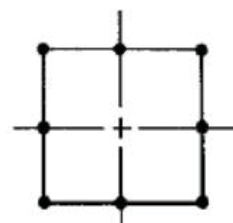
Rectangle



(1,3)

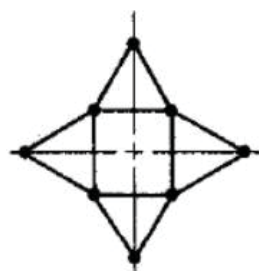


(1,7)

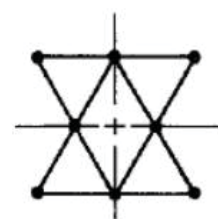


Rectangle

$M = 4$



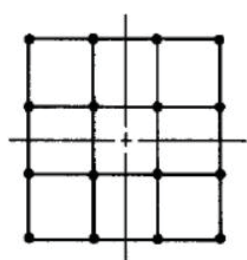
(4,4)



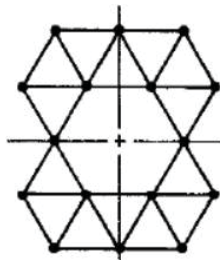
Triangle

$M = 8$

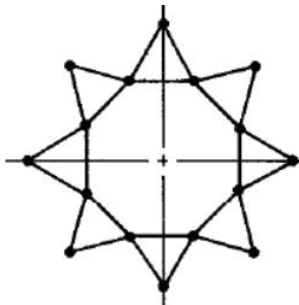
Various QAM Constellations (cont'd)



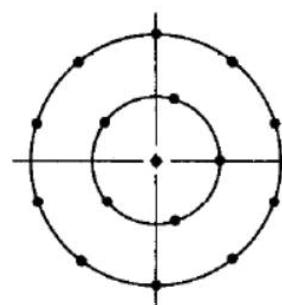
Rectangle



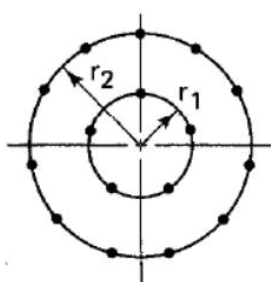
Triangle



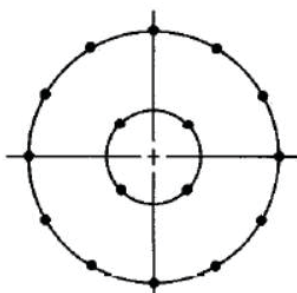
(8,8)



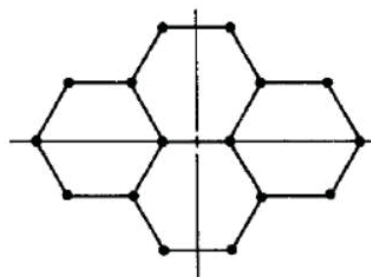
(1,5,10)



(5,11)



(4,12)



Hexagonal

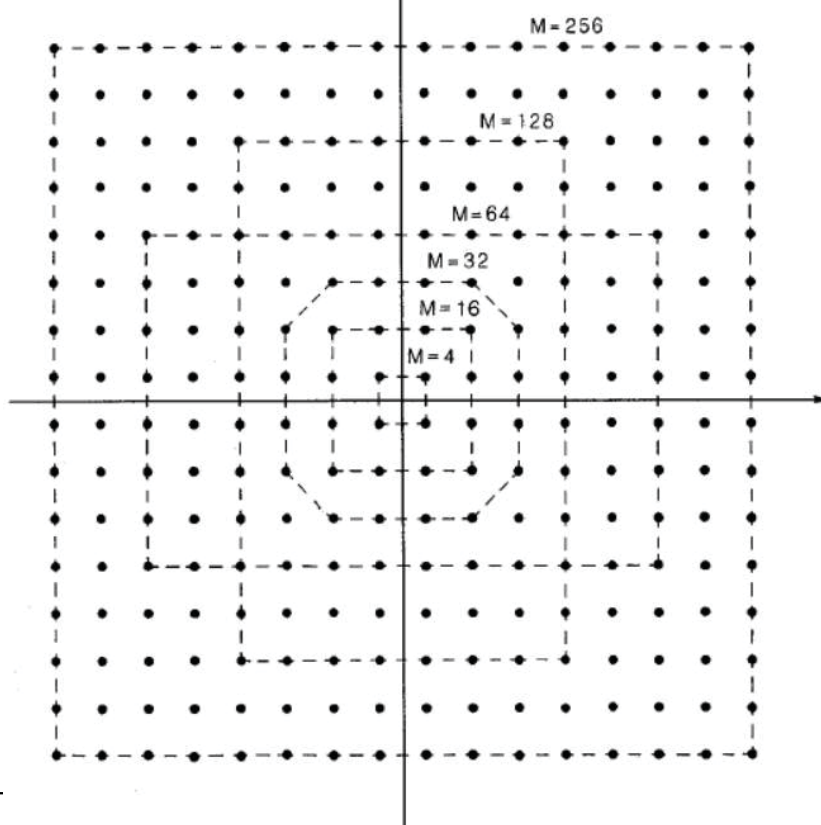
$M = 16$

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Square QAM Constellation



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Square QAM

- The signal model for M-ary square QAM is described as

$$s_m(t) = I_m \sqrt{\frac{E_0}{E_p}} \cdot p(t) \cos(w_c t) - Q_m \sqrt{\frac{E_0}{E_p}} \cdot p(t) \sin(w_c t)$$

$$= I_m \sqrt{\frac{E_0}{2}} \cdot \phi_1(t) - Q_m \sqrt{\frac{E_0}{2}} \cdot \phi_2(t)$$

- E_0 : the energy of the signal with lowest amplitude
- (I_m, Q_m) : the signal point in constellation
- The pair (I_m, Q_m) is an element of $L \times L$ matrix.

$$[I_m, Q_m] = \begin{bmatrix} (-L+1, L-1) & (-L+3, L-1) & \cdots & (L-3, L-1) \\ (-L+1, L-3) & (-L+3, L-3) & \cdots & (L-3, L-3) \\ \vdots & \vdots & \cdots & \vdots \\ (-L+1, -L+1) & (-L+3, -L+1) & \cdots & (L-3, -L+1) \end{bmatrix}$$

Square QAM (cont'd)

- $L = \sqrt{M}, M = 4^n, n = 1, 2, 3, \dots$
- L : positive integer
- For example, 16-QAM, $M = 16, L = \sqrt{16} = 4$

$$[I_m, Q_m]_{L \times L} = \begin{bmatrix} (-3, 3) & (-1, 3) & (1, 3) & (3, 3) \\ (-3, 1) & (-1, 1) & (1, 1) & (3, 1) \\ (-3, -1) & (-1, -1) & (1, -1) & (3, -1) \\ (-3, -3) & (-1, -3) & (1, -3) & (3, -3) \end{bmatrix}_{4 \times 4}$$

- For 32-QAM

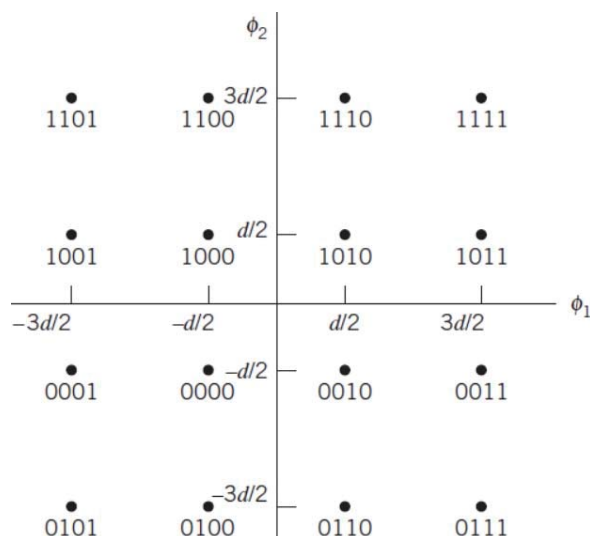
- $M = 32, L \neq \sqrt{M} \because M \neq 4^n$
- $M = 32 = 2^n, n = 5$
- 6×6 matrix without 4-element on the four corners

Square QAM (cont'd)

□ 16-QAM with Gray-code

□ Average power

■ P_{avg}



Square QAM (cont'd)

□ The magnitude of a phasor

■ $\|s_m\| = \sqrt{\frac{E_0}{2} (I_m^2 + Q_m^2)}$

□ The QAM signal can be reformulated as $s_m(t) = A_m \cos(\omega_c t + \theta_m)$

■ $A_m = \sqrt{\frac{E_0}{2} (I_m^2 + Q_m^2)} = \sqrt{\frac{E_0}{2}} \|s_m\|$

■ $\theta_m = \tan^{-1}(Q_m/I_m)$

■ The distance between any pair of phasor

• $d_{mn} = \sqrt{|s_m - s_n|^2} = \sqrt{\frac{E_0}{2} [(I_m - I_n)^2 + (Q_m - Q_n)^2]}$

• $m, n = 1, 2, \dots, M$

Square QAM (cont'd)

■ The average energy

- $E_{avg} = \mathbf{E} \left[\frac{E_0}{2} (I_m^2 + Q_m^2) \right] = \frac{E_0}{2} \{ \mathbf{E}[I_m^2] + \mathbf{E}[Q_m^2] \} = E_0 \mathbf{E}[I_m^2]$

■ The average power

- $P_{avg} = \frac{E_{avg}}{T} = \frac{E_0}{T} \mathbf{E}[I_m^2]$

- $$\begin{aligned} \mathbf{E}[I_m^2] &= \frac{1}{L} [(-L+1)^2 + (-L+3)^2 + \cdots + (L-3)^2 + (L-1)^2] \\ &= \frac{2}{L} [1^2 + 3^2 + \cdots + (L-1)^2] \\ &= \frac{1}{3} (L^2 - 1) = \frac{1}{3} (M - 1) \end{aligned}$$

- $P_{avg} = \frac{E_0}{3T} (M - 1) = P_0 \frac{1}{3} (M - 1)$

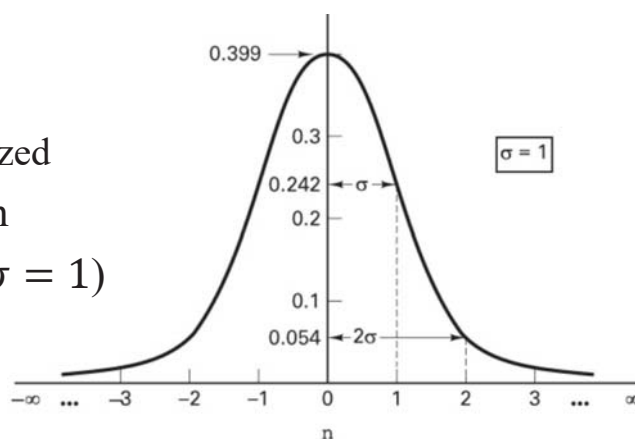
- P_0 : the power of the smallest signal

Probability of Error Due to Noise

□ Gaussian probability density function

- $p(n) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{n}{\sigma} \right)^2 \right]$

- σ^2 : the variance of n
- The normalized or standardized Gaussian density function of a zero-mean process ($\sigma = 1$)



Probability of Error Due to Noise (cont'd)

- A random signal $\mathbf{z}$ as the sum of a Gaussian noise random variable $\mathbf{n}$ and a dc signal $\mathbf{a}$ is given as

$$z = a + n$$

- The pdf $p(z)$ is expressed as

$$p(z) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{z - \mathbf{a}}{\sigma} \right)^2 \right]$$

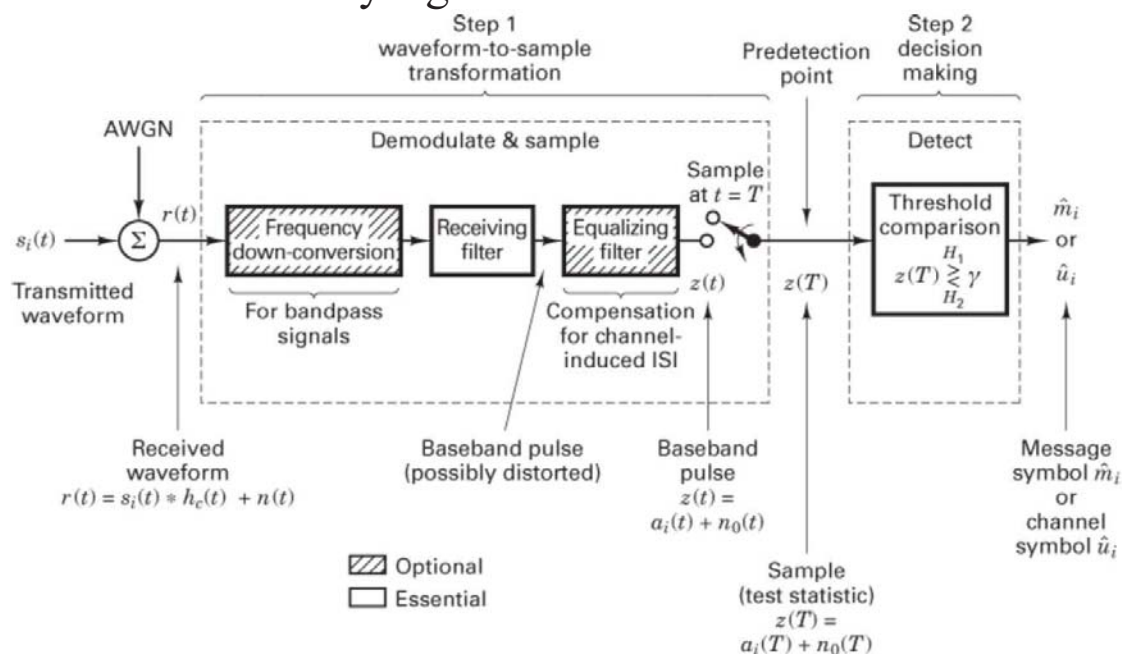
- σ^2 : the variance of n

- Recall the general form of Gaussian probability density function

$$p(n) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{n}{\sigma} \right)^2 \right]$$

Bit-Error Rate (BER)

- Detection of binary signal in Gaussian noise



Bit-Error Rate (cont'd)

- The transmitted signal over a symbol interval $(0, T)$ is represented as

- $s_i(t) = \begin{cases} s_1(t), & 0 \leq t \leq T, \text{ for a binary 1} \\ s_2(t), & 0 \leq t \leq T, \text{ for a binary 0} \end{cases}$

- The signal $r(t)$, received by the receiver, is given as

$$r(t) = s_i(t) * h_c(t) + n(t)$$

- Considering an ideal distortionless channel transmission, the $r(t)$ can be revised as

$$r(t) = s_i(t) + n(t), i = 1, 2; 0 \leq t \leq T$$

$n(t)$: zero-mean AWGN process (Gaussian distribution)

Note: $T = T_b$ because this example is binary transmission.

Bit-Error Rate (cont'd)

- First-step:

- Reduce the received signal waveform $r(t)$ to a single number $z(t = T)$

- The pre-detection point is given as

$$z(T) = a_i(T) + n_0(T), i = 1, 2;$$

$a_i(T)$: the signal component of $z(T)$

$n_0(T)$: the noise component (zero-mean Gaussian random variable)

$z(T)$: the Gaussian random variable with a mean of either a_1 or a_2

- The pdf of the Gaussian random noise n_0

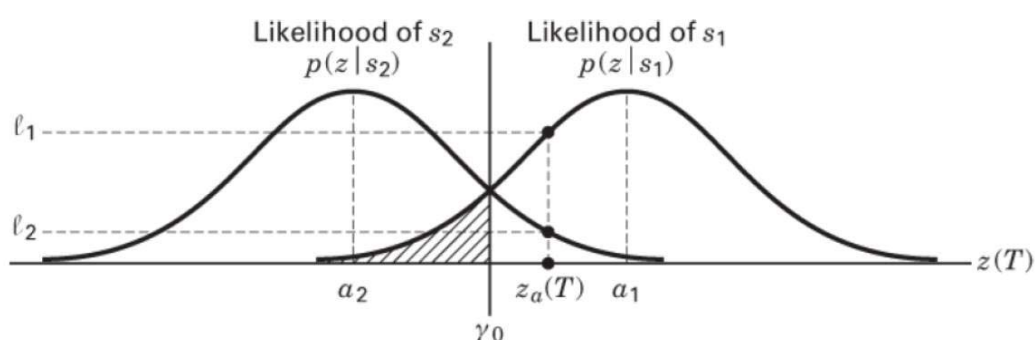
$$p(n_0) = \frac{1}{\sigma_0 \sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{n_0}{\sigma_0} \right)^2 \right]$$

Bit-Error Rate (cont'd)

- The conditional probability density functions (pdfs) $p(z|s_1)$ and $p(z|s_2)$ can be expressed as

$$p(z|s_1) = \frac{1}{\sigma_0\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{z - a_1}{\sigma_0} \right)^2 \right]$$

$$p(z|s_2) = \frac{1}{\sigma_0\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{z - a_2}{\sigma_0} \right)^2 \right]$$



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Bit-Error Rate (cont'd)

□ Second-step:

- Signal detection based on comparing the test statistic $z(T)$ to a threshold level γ
- The signal energy (not its shape) is the important parameter in the detection process.
- Detection rule

$$\begin{array}{c} H_1 \\ z(T) > \gamma \\ H_2 \end{array}$$

- H_1 and H_2 : two possible (binary) hypotheses; H_1 (H_2) is equivalent to decide which signal s_1 (s_2) was sent.
- Hypothesis H_1 is chosen if $z(T) > \gamma$, and hypothesis H_2 is chosen if $z(T) < \gamma$.

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Bit-Error Rate (cont'd)

- Maximum likelihood (ML) receiver structure
 - Minimize the probability of error based on choosing the threshold level γ for the binary decision
 - $\gamma = \gamma_0$: the minimum error value

- The likelihood ratio test is described as

$$\frac{p(z|s_1)}{p(z|s_2)} \underset{H_2}{\overset{H_1}{>}} \frac{P(s_2)}{P(s_1)}$$

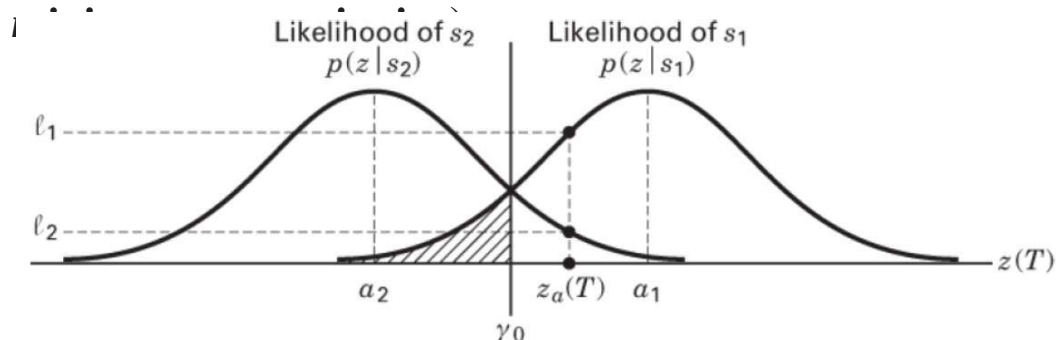
- $P(s_1)$ and $P(s_2)$: a prior probabilities of s_1 and s_2

Bit-Error Rate (cont'd)

- If $P(s_1) = P(s_2)$ and the likelihoods are symmetrical $\{p(z|s_1) = p(z|s_2)\}$, the decision rule can be revised as

$$z(T) \underset{H_2}{\overset{H_1}{>}} \frac{a_1 + a_2}{2} = \gamma_0$$

- γ_0 : the **optimum threshold (the decision boundary)** for minimizing the probability of making an incorrect decision (the



Bit-Error Rate (cont'd)

□ Error probability

- An error e will occur when s_1 is sent. Then the probability of error is given as

$$p(e|s_1) = P(H_2|s_1) = \int_{-\infty}^{\gamma_0} p(z|s_1) dz$$

- Similarly, an error e will occur when s_2 is sent. Then the probability of error is given as

$$p(e|s_2) = P(H_1|s_2) = \int_{\gamma_0}^{\infty} p(z|s_2) dz$$

- For the binary case, the probability of bit error P_B is derived as

$$\begin{aligned} P_B &= \sum_{i=1}^2 p(e|s_i) = p(e|s_1)P(s_1) + p(e|s_2)P(s_2) \\ &= P(H_2|s_1)P(s_1) + P(H_1|s_2)P(s_2) \end{aligned}$$

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Bit-Error Rate (cont'd)

- The prior probabilities are equivalent, i.e., $P(s_1) = P(s_2) = \frac{1}{2}$.

$$P_B = P(H_2|s_1)P(s_1) + P(H_1|s_2)P(s_2) = \frac{1}{2}P(H_2|s_1) + \frac{1}{2}P(H_1|s_2)$$

where $P(H_2|s_1) = P(H_1|s_2)$. Then, we have

$$P_B = P(H_2|s_1) = P(H_1|s_2)$$

- Compute P_B by integrating $p(z|s_1)$ between the limits $-\infty$ and γ_0 (or integrating $p(z|s_2)$ between the limits γ_0 and ∞)

$$P_B = \int_{\gamma_0}^{\infty} p(z|s_2) dz = \int_{\gamma_0 = \frac{a_1+a_2}{2}}^{\infty} \frac{1}{\sigma_0\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{z-a_2}{\sigma_0}\right)^2\right] dz$$

- Let $u = (z - a_2)/\sigma_0$ and $\sigma_0 du = dz$

$$P_B = \int_{u = \frac{a_1-a_2}{2\sigma_0}}^{u=\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}u^2\right] du = Q\left(\frac{a_1 - a_2}{2\sigma_0}\right)$$

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Bit-Error Rate (cont'd)

- $Q(x)$: the complementary error function or co-error function
 - It is common used symbol for the probability under the tail of the Gaussian distribution.

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} \exp\left[-\frac{1}{2}u^2\right] du$$

- Approximate to $Q(x)$ by simpler function

$$Q(x) = \frac{1}{x\sqrt{2\pi}} \exp\left[-\frac{1}{2}x^2\right], \text{ for } x > 3$$

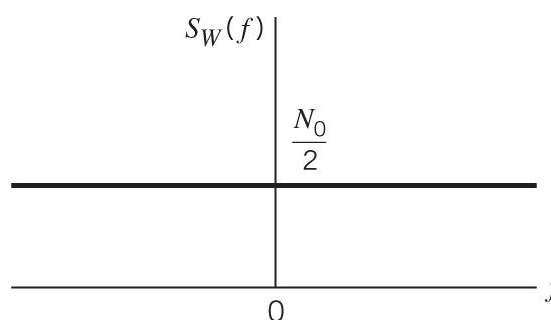
$$Q(x) = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right)$$

- White noise

$$\sigma_0^2 = \frac{N_0}{2} \text{ and } N_0 = kT_e$$

k : Boltzmann's constant

T_e : the equivalent noise temperature of receiver



Bit-Error Rate (cont'd)

- Complementary error function

$$\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^{\infty} \exp[-u^2] du$$

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} \exp\left[-\frac{1}{2}u^2\right] du$$

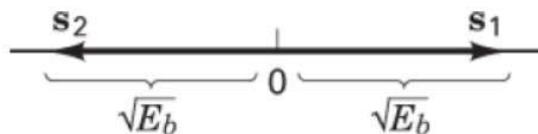
- Relationship between $Q(\cdot)$ and $\operatorname{erfc}(\cdot)$

$$Q(x) = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right)$$

Bit-Error Rate (cont'd)

□ BPSK

- $a_1 = \sqrt{E_b}$ and $a_2 = -\sqrt{E_b}$
- E_b : the signal energy per binary symbol
- $\sigma_0^2 = \frac{N_0}{2}$
- BER



$$P_B = \int_{u=\frac{a_1-a_2}{2\sigma_0}}^{u=\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}u^2\right] du = \int_{\sqrt{\frac{2E_b}{N_0}}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}u^2\right] du$$

$$= Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

- Recall $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} \exp\left[-\frac{1}{2}u^2\right] du$ and $P_B = Q\left(\frac{a_1-a_2}{2\sigma_0}\right)$

Bit-Error Rate (cont'd)

□ Basic SNR parameter for digital communication systems

- SNR: the ratio of average signal power to average noise power
- $SNR = S/N$
 - S : the average modulating signal power
 - N : the average noise power

□ E_b/N_0 can be expressed in terms of SNR , W (Bandwidth), T (bit time duration), and R (bit rate).

$$\frac{E_b}{N_0} = \frac{ST}{N_0} = \frac{S}{RN_0} = \frac{SW}{RN_0W} = \frac{S}{N} \left(\frac{W}{R}\right)$$

- $R = 1/T$
- $N = N_0W$

Bit-Error Rate (cont'd)

□ For example

- Find P_B for a BPSK system with a bit rate of 1 Mbit/s. The received waveforms, $s_1(t) = A \cos w_0 t$ and $s_2(t) = -A \cos w_0 t$, are coherently detected with a matched filter. The value of A is 10 mV. Assume the single-sided noise power spectral density is $N_0 = 10^{-11} \text{ W/Hz}$ and the signal power and energy per bit are normalized relative to a 1Ω load.

■ Solution:

$$A = \sqrt{\frac{2E_b}{T}} = 10^{-2} \text{ V}, T = \frac{1}{R} = 10^{-6} \text{ s}$$

$$E_b = \frac{A^2}{2} T = 5 \times 10^{-11} \text{ J} \text{ and } \sqrt{\frac{2E_b}{N_0}} = 3.16$$

$$P_B = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = Q(3.16) = 8 \times 10^{-4}$$

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Shannon – Channel Capacity Theorem

- The system capacity, C , of a channel perturbed by additive white Gaussian noise (AWGN) is a function of the average received signal power, S , the average noise power, N , and the BW, W .

- $C = W \log_2 \left(1 + \frac{S}{N}\right)$ (bits/s)

- W (Hz)
- $R \leq C$, R where is transmission rate.

- A limit on transmission rate, not on error probability.

□ Normalized channel capacity

- $\frac{C}{W}$ (bits/s/Hz) is defined as a function of SNR in the channel.

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Shannon – Channel Capacity Theorem (cont'd)

- The detected noise power is proportional to BW.

- $N = N_0 W$

- The channel capacity C can be reformulated as

- $\frac{C}{W} = \log_2 \left(1 + \frac{S}{N_0 W} \right)$ (bits/s)

- For the case, the transmission bit rate is equal to channel capacity, $R = C$,

- Considering the transmission bit rate is equivalent to channel capacity, namely $R = C$, we can use the following equation.

- $\frac{E_b}{N_0} = \frac{S}{RN_0} = \frac{S}{N_0 C}$

- The channel capacity can be revised as

- $\frac{C}{W} = \log_2 \left(1 + \frac{E_b}{N_0} \frac{C}{W} \right)$